Additional file 1: Supplementary methods

What Does EEG-to-fMRI Synthesis Add to EEG-Based Diagnosis? A Controlled Benchmark in Psychosis, Schizophrenia and Parkinson's Disease

David Calhas and Rui Henriques

S1. Datasets and data checks

Table 1 of the main text summarises the datasets. All are previously published, anonymised public releases analysed here as secondary data.

First-episode psychosis (FEP1, FEP2). Two independently collected samples from the University of Pittsburgh (OpenNeuro ds003944 and ds003947) [31, 32]: individuals with a first episode of psychosis and healthy controls; five minutes of eyes-open resting EEG recorded with a 10-10 cap inside an Elekta Neuromag Vectorview MEG system (61 EEG channels, 1000 Hz, reference TP9). Age, sex, symptom and functioning scales (BPRS, SANS, SAPS, GAF, SFS), cognition (MATRICS, WASI) and chlorpromazine-equivalent medication are released. FEP1 lists 82 recordings, but five people appear twice under the identifiers `sub-X` and `sub-XA` (same age, sex and race, recorded days apart); we kept one recording per person (47 patients, 30 controls). FEP2 contains 31 patients and 30 controls and excludes, by construction, the subjects shared with FEP1.

Chronic schizophrenia (Semmelweis). Resting EEG from Semmelweis University, Budapest [27]: 38 patients with schizophrenia and 39 age- and sex-matched controls; about two minutes of eyes-closed rest, 64-channel Neuroscan amplifier at 1000 Hz. Patients are identified by the presence of PANSS ratings, which reproduces the 38/39 split of the source publication (35 of the 38 carry an ICD-10 F20/F25 code in the released table; three have a missing or non-F2 code). Medication, illness duration, education and PANSS scores are released.

Parkinson's disease (PD). Resting EEG from the University of New Mexico (OpenNeuro ds003490) [8]: 25 patients with Parkinson's disease and 25 matched controls; 63 EEG channels, 500 Hz (Brain Vision ActiChamp; reference CPz), instructed one-minute eyes-closed and eyes-open spans, followed in the same file by an auditory oddball task (Section S3), plus EOG and a three-axis accelerometer on the most tremor-affected hand. Patients were recorded twice, a week apart, ON and OFF dopaminergic medication (13 ON first, 12 OFF first); we analysed the OFF session. Eight subjects of this cohort were used by [37].

Schizophrenia spectrum, task EEG (Fribourg). The cohort used in [6], released by [25]: 19 patients with a psychotic disorder (ICD-10) and 24 controls, recorded during the Ultimatum Game (responder role), 128-channel Biosemi ActiveTwo at 2048 Hz; the release includes additional EXG channels whose layout differs between groups (four in all controls, six in 18 of 19 patients), which are excluded by the channel selection of Section S3. The public release distributes the group label only; group-level characteristics are reported in [15].

Data checks. For every cohort we checked for repeated participants, verified group sizes against the source publications, and tested whether recording metadata (channel set, sampling rate, filters, duration, recording date and hour, file size) differs between groups; fixed-length crops remove duration differences.

No recording metadata differed between groups in FEP1 or Parkinson's disease; in FEP2 the raw recordings of patients were longer (removed by the fixed crop). Semmelweis patients were recorded earlier in the day than controls (mean hour 14.0 against 16.3, $p < 0.001$), and in Fribourg the nine participants recorded in 2016 were all patients and the auxiliary channel layout differed by group (Additional file 2). Artefact burden differed by group in two cohorts: in Parkinson's disease controls had larger EEG amplitudes and more ocular activity than patients (EOG power, AUC 0.26, $p = 0.004$), consistent with the reduced blink rate of Parkinson's disease, whereas accelerometer power did not differ ($p = 0.28$); in Semmelweis patients had more high-amplitude segments ($p = 0.003$). The number of ocular components removed by ICA did not differ between groups in any cohort (all $p \geq 0.07$). In Parkinson's disease every recording followed the same order (one minute of eyes closed, then one minute of eyes open, then the oddball task), and the onsets of the rest spans and of the task did not differ between groups ($p = 0.54$ and $p = 0.31$); the fixed 0–400 s crop of the primary analysis therefore included a variable set-up period before rest and the start of the oddball task.

S2. Pairing EEG and fMRI, and synthesizer training

fMRI is available only in the synthesis training dataset, NODDI [9, 10]; the clinical cohorts are EEG-only and contain no fMRI. In NODDI, fMRI was acquired simultaneously with EEG at rest (eyes open, fixation) on a 1.5 T Siemens Avanto: T2*-weighted gradient-echo EPI, TR/TE 2160/30 ms, 30 slices of 3 mm (1 mm gap), effective voxel size $3.3 \times 3.3 \times 4.0$ mm, $64 \times 64 \times 30$ voxels, 300 volumes; EEG with a 64-channel MR-compatible cap (BrainCap MR), recorded at 5 kHz, released at 250 Hz after artefact correction.

Two properties of this release matter for synthesis and, to our knowledge, have not been reported. First, in 15 of the 16 subjects the EEG begins 13–130 s before the first fMRI volume; pairing EEG and fMRI from the start of the file would pair EEG with volumes 6–60 TRs away. We pair them through the scanner volume markers in the EEG file (one per TR). Second, two electrode montages are present: 10 subjects have F1/F2/C1/C2/P1/P2 and 6 have P9/P10/PO9/PO10/O9/O10, in a different channel order, so positional channel indexing mixes electrodes across subjects. We use the 57 channels recorded in all subjects, selected by name; the ECG channel is excluded.

Synthesizer training. The synthesis target is the raw EPI volume min–max scaled to $[0, 1]$ per volume, the convention of the published volumetric synthesizers. Synthesizers used for classification were trained on 14 NODDI subjects, with two subjects held out to select the training epoch; loss $0.5(1-\text{SSIM})+0.5\,\text{MSE}$, Adam, learning rate 10^{-3}, batch size 32, mini-batches shuffled across subjects, 50 epochs (15 for Spec2VolCAMU-Net). NT-ViT used the authors' recipe (AdamW, learning rate 5×10^{-4}, 30 epochs) with batch size 32 instead of 128, because the 43.2 s input did not fit in memory at 128. Input features were standardised with training-subject statistics only. The epoch with the highest validation SSIM was kept: epoch 11 for E2fNet, 10 for Spec2VolCAMU-Net and 20 for NT-ViT, but the first epoch (one pass over the training volumes) for CNN-TAG, whose validation SSIM did not improve afterwards (the same happened in 83 of its 96 leave-one-subject-out runs). Because a model selected after one epoch is close to its initialisation, CNN-TAG was also trained for all 50 epochs without selection and evaluated as a sensitivity analysis. NODDI carries no diagnostic labels and is never used for classification.

Second training dataset (XP1). To test whether the results depend on the training data, all four synthesizers were retrained on a second simultaneous EEG-fMRI dataset, XP1 (OpenNeuro ds002336) [23]: ten healthy adults, 3 T, TR 2 s, 32 slices of 4 mm, 64-channel MR-compatible EEG (BrainCap MR, the same layout as one NODDI montage, so the 57 channels are present by name), four runs each of motor imagery, neurofeedback and a motor localiser. We used the released gradient- and pulse-corrected EEG and matched each of its volume markers to a scanner trigger of the raw recording, anchored on the task markers present in both and validated on all of them. Because the number of raw triggers differed from the number of volumes in 15 of the 35 runs (one extra or up to six missing), only the 19 runs with exactly one trigger per volume were used (all ten subjects, about 3,700 volumes). Volumes were resampled from $106 \times 106 \times 32$ to the $64 \times 64 \times 30$ grid of NODDI, and EEG went through the same harmonisation, so the clinical inputs are unchanged. Eight subjects were used for training and two to select the epoch, with the NODDI training settings.

S3. EEG harmonisation and artefact handling

EEG datasets differ in electrode count and layout, sampling rate, reference and paradigm, which changes their representation in $\mathbb{R}^{C \times T}$. We apply a single harmonisation to every recording, clinical or not, so that each synthesizer receives inputs in exactly the montage and sampling rate it was trained on:

Channel selection by type. EEG channels are selected by their declared type, never by position; EOG, ECG, EMG, accelerometer, trigger and auxiliary (EXG) channels are removed. Selection by type also keeps acquisition differences unrelated to the disease out of the input: in the Fribourg release, for example, the auxiliary (EXG) channel layout differs between patients and controls.

Montage harmonisation. Channels are mapped to standard 10-05 positions and interpolated with spherical splines onto the 57 EEG channels recorded in every NODDI subject (Section S2).

Common filtering and resampling. Notch filtering at the mains frequency and harmonics, 0.5–45 Hz band-pass, average reference and resampling to 250 Hz. Frequencies above 45 Hz are discarded for every dataset, so all synthesizers see the same frequency range.

Clinical EEG has no fMRI volume markers; the synthesizers are applied to consecutive pseudo-TRs of 2.16 s starting 43.2 s into each recording, so that every synthesized volume has a full 20-segment history.

All recordings went through this harmonisation (MNE-Python). Each clinical recording was cropped to a fixed duration per cohort (Table 1 of the main text), identical for patients and controls, to remove duration differences between groups. The cleaned NODDI EEG released by the dataset authors, in which scanner (gradient) and pulse (ballistocardiographic) artefacts were removed [10], was used as is before harmonisation.

Artefact handling. The primary analysis applies no artefact rejection or independent component analysis (ICA), so that every view receives the same input. As a sensitivity analysis, every recording was re-harmonised with ICA (FastICA, 20 components fitted on a 1 Hz high-passed copy, before interpolation), removing the components correlated with the EOG (MNE `find_bads_eog`, $z > 3$), using only EOG channels present in every recording of a cohort so that the cleaning could not differ by group through the channel set; all views, synthesized ones included, were then recomputed. The standard muscle criterion (`find_bads_muscle`) flagged most components of the 45 Hz low-passed data and was not used for removal; muscle artefacts were instead addressed by recomputing band power without the 30–45 Hz band. Artefact burden was compared between groups through the fraction of pseudo-TRs exceeding 150 μV peak-to-peak or an EMG-like 30–45 Hz power (median plus three robust standard deviations), the median peak-to-peak amplitude, the power of the EOG channels and, in Parkinson's disease, the power of the accelerometer on the most affected hand. In Parkinson's disease, the released rest file contains a set-up period, one minute of eyes closed, one minute of eyes open and then an auditory oddball task; besides the fixed 0–400 s crop of the primary analysis, a rest-only crop from the first to the last eyes-closed or eyes-open marker was analysed, with and without ICA.

S4. EEG views and synthesizers

Each subject's recording, $\vec{x}_0 \in \mathbb{R}^{D_0}$, is labelled $y \in \{0, 1\}$ (control or patient) and represented by views $v_i = F_i(\vec{x}_0) \in \mathbb{R}^{D_i}$, where F_i is a fixed or learned transformation with parameters θ_i: the band-power view v_0, the STFT view v_1 and the synthesized-fMRI view v_2, defined below. A view is evaluated by an l_2-regularised logistic regression minimising

$$\mathcal{L}_C = -\sum_s \left[y_s \log \hat{p}_s + (1 - y_s) \log(1 - \hat{p}_s)\right] + \frac{1}{2C}\|\vec{w}\|_2^2, \qquad \hat{p}_s = \sigma\left(\vec{w}^\top v_i(\vec{x}_{0,s}) + b\right), \tag{S1}$$

over training subjects s, where $\vec{w} \in \mathbb{R}^{D_i}$ and $b \in \mathbb{R}$ are the classifier parameters, σ is the logistic function and C is the inverse regularisation strength. The question is whether a synthesized-fMRI view, alone or added to an EEG view, lowers the expected loss on unseen subjects relative to the EEG view alone.

0.0.1 Band-power view

Frequency-domain EEG features are long-established markers in schizophrenia and Parkinson's disease [3, 4, 16, 29]. We use the simplest such view: for each of the $C = 57$ channels (Section S3), Welch power spectra (4 s windows) summarised as relative log power in five bands (1–4, 4–8, 8–13, 13–30, 30–45 Hz), so $v_0(\vec{x}_0) \in \mathbb{R}^{C \times 5}$. Band power is spatially organised over the scalp and clinically interpretable, so it is both the natural EEG-only baseline and the information any synthesized view starts from.

0.0.2 STFT view

Time-frequency features are widely used in EEG studies of psychiatric and neurological disorders [1, 4, 5, 11, 29]. The synthesizers take a short-time Fourier representation [17] as input, $\vec{x}_1 = v_1(\vec{x}_0) \in \mathbb{R}^{D_1}$, $D_1 = C \times F \times T$: log-magnitude spectra of $T = 20$ consecutive EEG segments of one fMRI repetition time (TR, 2.16 s) each, between 0.46 and 45 Hz ($F = 97$ bins), i.e. the 43.2 s of EEG preceding an fMRI volume, which covers the haemodynamic delay. This view is the input of every synthesizer. Its temporal mean and standard deviation over a recording's pseudo-TRs (Section S3) also give an EEG-only view that is matched in information to the synthesized-fMRI view (Section S6).

The first synthesizer is the network introduced and validated in our earlier work [7]; its architecture is adopted from that study, not proposed here, and the present work re-implements it in PyTorch and retrains it with the protocol of Section S2. We refer to it as CNN-TAG. Its flow is illustrated in Figure S1. Its input is the STFT view, $\vec{x}_1$; its output is the synthesized-fMRI view, $\vec{x}_2 \in \mathbb{R}^{M_1 \times M_2 \times M_3}$, a volume of $64 \times 64 \times 30$ voxels. The synthesized-fMRI view is therefore the composition $v_2 = F \circ v_1$ of the STFT transform and the network F.

In terms of structure, the network processes the input with an attention mechanism over channels. An attention weight matrix $A \in \mathbb{R}^{C \times (FT)}$ is used to compute $E \in \mathbb{R}^{C \times C}$, $E_{ij} = \vec{x}_i^\top A_j$, which is normalised with sparsemax along j to give B; the attention-processed representation is

$$T_i = \sum_j^C b_{ij}\, \vec{x}_j, \forall i \in \{1, \ldots, C\}. \tag{S2}$$

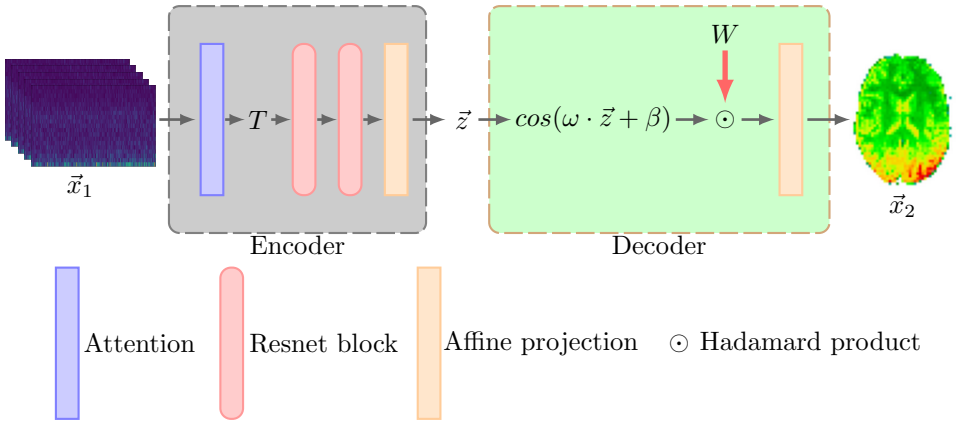


Figure S1: **Architecture of the CNN-TAG synthesizer.** The neural architecture has two components: an Encoder (shaded in grey) and a Decoder (shaded in green). The input is the STFT v_1 representation. The output is the synthesized fMRI. The Encoder begins with a simple attention mechanism on the channels dimension of the STFT. After, it is processed by two Resnet blocks and an affine layer. This produces the latent representation $\vec{z}$. Following, the Decoder picks this representation and builds the cosine bases through the projection $\omega \cdot \vec{z} + \beta$. The sinusoids are style induced with $cos(\omega \cdot \vec{z} + \beta) \odot W$. The vector W is a style fixed pretrained vector of an fMRI representation, learned from a simultaneous EEG and fMRI dataset (described in Section S2). Finally, an affine layer projects it to the fMRI space.

T is processed by two residual blocks [14] and a fully connected layer, whose parameters θ_E form the encoder. The encoder output projects random Fourier features [28, 34], parameterized by ω and β and layer-normalised around $\pi/2$; the resulting sinusoids are multiplied with a learned latent style vector, $W \in \mathbb{R}^L$, and an affine transformation maps them to the fMRI volume space. With this, we formulate the function structure $F : \mathbb{R}^{C \times F \times T} \rightarrow \mathbb{R}^{D_2}$.

Other synthesizers. Three published synthesizers were trained with the same data and protocol, using the authors' model code: **E2fNet** [30], a convolutional EEG encoder followed by a 2D U-Net and a convolutional fMRI decoder; **Spec2VolCAMU-Net** [13], a multi-scale convolutional spectrogram encoder with temporal self-attention and a Vision-Mamba U-Net decoder, whose selective scan we computed with an exact pure-PyTorch chunked recurrence because the CUDA kernel was unavailable; and **NT-ViT** [19], a mel-spectrogram vision-transformer encoder and decoder with a domain-matching loss, which computes its own spectrograms and therefore received the same 43.2 s window as raw waveforms (the published model used a simultaneous 2.16 s window without haemodynamic lag). CATD [37] could not be included because no code is available; NeuroBOLT and UnEBOLT [21, 22] predict parcel time series rather than volumes and were used in the supporting fidelity analyses only.

Untrained controls. For each synthesizer, the same architecture with random initial weights was passed through the identical pipeline. A fixed nonlinear projection of EEG spectra into a voxel grid can support classification on its own; this control separates that effect from the learned EEG-to-fMRI mapping.

From synthesized volumes to a subject view. For each clinical subject, the synthesized volumes (one per pseudo-TR, clipped to $[0, 1]$) are summarised by their voxelwise temporal mean and standard deviation, giving $2 \times 64 \times 64 \times 30$ features; this is the synthesized-fMRI view classified in Section S6.

S5. Contrastive sinusoid separation

In addition to the four synthesizers, we evaluate the integration approach of the earlier version of this work, which adapts the CNN-TAG synthesizer to the diagnostic task by separating classes in its sinusoid space while the voxel decoder is kept fixed.

CNN-TAG passes its encoder output $\vec{z}$ through random Fourier features, $\cos(\omega \cdot \vec{z} + \beta)$. Because the cosine is periodic, classes separated in $\vec{z}$ can overlap after the projection. The method therefore layer-normalises [2] $\vec{z}$ around $\pi/2$ with standard deviation $\pi/2$, which places most projections in an interval where the cosine is monotonic, and trains the encoder with the contrastive loss

$$\mathcal{L}_D(p_1, p_2, y_p) = y_p \, D(p_1, p_2) + (1 - y_p) \, |D(p_1, p_2) - m| \,, \tag{S3}$$

where p_1, p_2 formulate a pair of two instances and $y_p \in \{0, 1\}$ defines the pairwise label, being 0 when the

label of p_1, p_2 mismatch and 1 when they match. $D(p_1, p_2) = \|p_1 - p_2\|_1$ is the l_1 distance between the latent vectors of the pair, and $m \in (0, \pi]$ is the margin. Mismatched pairs are pulled to distance m by an absolute deviation rather than pushed beyond it by a hinge, $\max(0, m - D)$, as in the original formulation: with the layer normalisation, this keeps the latent coordinates of both classes inside one non-periodic interval of the cosine, whereas a hinge lets distances grow past π, where the cosine wraps and the classes overlap again after projection (Fig. S3). Under the random projection, the expected similarity of two projected points decreases exponentially with their latent distance, so a margin that separates the classes in $\vec{z}$ also separates them after projection; the statement, its proof and an illustration of the periodicity problem are given below. Figure S2 illustrates the effect before and after training.

Training. The encoder is trained with $\mathcal{L}_D$ on pairs of EEG windows, while a linear classifier with an l_1 penalty is trained with $\mathcal{L}_C$ on the synthesized volume (detached from the encoder); the style vector and the voxel decoder stay frozen at their NODDI weights. Each synthesized window is an instance, and a subject's score is the mean predicted probability over its windows. We run two variants: *re-initialised*, in which the encoder and Fourier projection start from random weights while the decoder keeps its NODDI weights, as in the original formulation, and *pretrained*, in which they start from the NODDI-trained synthesizer and are fine-tuned. Training uses a fixed budget (10 epochs, Adam, learning rate 10^{-3}, batches of 16 pairs, l_1 weight 10^{-4}), without per-fold tuning.

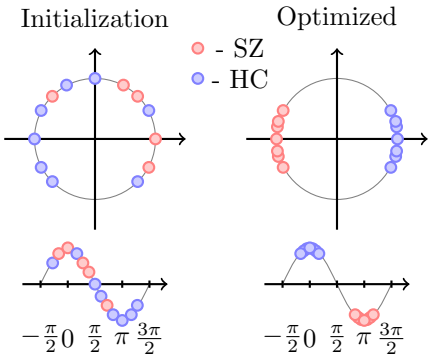


Figure S2: **Contrastive separation of sinusoids.** Normalization of data points inside the unit circle, using layer normalization, along with the optimization of a contrastive loss lead to correct separation of sinusoids. Data points belong to two classes, HC and SZ, that are separated after the minimization of $\mathcal{L}_D$. Because we separate false pairs, according to $(1 - y_p) \times ||D(x_1, x_2) - m||_1$, all points are placed within a shift variant interval of the cosine. This preserves the between-class variance that the classifier needs.

Periodicity of the cosine projection

The CNN-TAG synthesizer maps its encoder output $\vec{z}$ through random Fourier features, $\cos(\omega \cdot \vec{z} + \beta)$. In order to separate data along the projection of cosines, the method has to operate in a sub-domain interval where the cosine is not periodic. Layer normalization, with centre $\frac{\pi}{2}$ and standard deviation $\frac{\pi}{2}$, maps most of the data to such an interval of the cosine (Fig. S3). This alone does not separate the classes; the contrastive loss of the main text (Methods, contrastive sinusoid separation) moves pairs of different classes to a margin m apart in this interval.

Separation under the random cosine projection

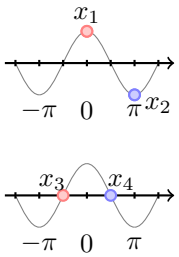
Proposition S1 (Separation under cosine projection)**.** *Let $z_1, z_2 \in \mathbb{R}^d$ and let*

$$\phi(z) = \left[\cos(\omega_1^\top z + \beta_1), \ldots, \cos(\omega_d^\top z + \beta_d)\right]$$

with independent $\omega_i \sim \mathcal{N}(0, I_d)$ and $\beta_i \sim \text{Uniform}[0, 2\pi]$. Then

$$\mathbb{E}_{\omega,\beta}\left[\phi(z_1)^\top \phi(z_2)\right] = \frac{d}{2} e^{-\frac{1}{2}\|z_1 - z_2\|_2^2},$$

so the expected similarity of the projected points decreases exponentially with their latent distance, and a contrastive margin that increases the latent distance between classes decreases their expected similarity after projection.



$$x_1, x_2 \sim \mathcal{X}_1 : ||x_1 - x_2||_1 = \pi$$
$$x \sim \mathcal{X}_1 : \mathbb{E}[x] = \tfrac{\pi}{2} \wedge \text{Var}[x] = \tfrac{\pi}{2}$$
$$x_3, x_4 \sim \mathcal{X}_2 : ||x_3 - x_4||_1 = \pi$$
$$x \sim \mathcal{X}_2 : \mathbb{E}[x] = 0 \wedge \text{Var}[x] = \tfrac{\pi}{2}$$

Figure S3: Description of how two similarly distributed samples, taken from $\mathcal{X}_1$ and $\mathcal{X}_2$, can lead to different portions of the cosine function image, since a sinusoid is periodic. Two distributions $\mathcal{X}_1$ and $\mathcal{X}_2$, may be mapped to the same image (second/bottom example). This is why a cosine is a shift invariant function. However, there are intervals a shift can be made and it is not invariant. Such intervals take the form $\forall i \in \mathbb{Z} : [i\pi, (i+1)\pi]$.

Proof of Proposition S1. Let

$$\phi(z) = \left[\cos(\omega_1^\top z + \beta_1), \ldots, \cos(\omega_d^\top z + \beta_d)\right],$$

where each $\omega_i \sim \mathcal{N}(0, I_d)$ and $\beta_i \sim \text{Uniform}[0, 2\pi]$. The inner product between projected vectors is:

$$\phi(z_1)^\top \phi(z_2) = \sum_{i=1}^{d} \cos(\omega_i^\top z_1 + \beta_i) \cos(\omega_i^\top z_2 + \beta_i).$$

Using the trigonometric identity:

$$\cos(a)\cos(b) = \frac{1}{2}[\cos(a-b) + \cos(a+b)],$$

we write:

$$\cos(\omega_i^\top z_1 + \beta_i) \cos(\omega_i^\top z_2 + \beta_i) =$$
$$\frac{1}{2}\left[\cos(\omega_i^\top (z_1 - z_2)) + \cos(\omega_i^\top (z_1 + z_2) + 2\beta_i)\right].$$

Taking the expectation over $\beta_i \sim \text{Uniform}[0, 2\pi]$, the second term vanishes:

$$\mathbb{E}_{\beta_i}[\cos(\omega_i^\top (z_1 + z_2) + 2\beta_i)] = 0.$$

Thus,

$$\mathbb{E}_{\omega,\beta}[\phi(z_1)^\top \phi(z_2)] = \frac{1}{2} \sum_{i=1}^{d} \mathbb{E}_{\omega_i}[\cos(\omega_i^\top (z_1 - z_2))].$$

Let $\delta = z_1 - z_2$. Since $\omega_i^\top \delta \sim \mathcal{N}(0, \|\delta\|_2^2)$, we use the identity:

$$\mathbb{E}_{x \sim \mathcal{N}(0,\sigma^2)}[\cos(x)] = e^{-\frac{1}{2}\sigma^2},$$

to obtain:

$$\mathbb{E}_{\omega_i}[\cos(\omega_i^\top \delta)] = e^{-\frac{1}{2}\|\delta\|_2^2}.$$

Putting it together:

$$\mathbb{E}_{\omega,\beta}[\phi(z_1)^\top \phi(z_2)] = \frac{d}{2} e^{-\frac{1}{2}\|z_1 - z_2\|_2^2}.$$

which is at most $d/2$ and decreases strictly and exponentially with $\|z_1 - z_2\|_2$, proving the result. □

Extension to more than two classes

A multi-class solution is to consider the unitary circle in Fig. S4. Let the pairs of data be $(x_1, x_2, y_1, y_2, y_p)$. By definition $x_1 := \mathbf{x}_2 \in \mathbb{R}^E$. And consider the targets $y_1, y_2 \in [0, 1]^C$. Now y_p instead of being in $[0, 1]$, consider $y_p = ||argmax(y_1) - argmax(y_2)||_1^1 \times \frac{\pi}{C}$ so that the possible values are $\in \{0, \frac{\pi}{C}, \ldots, \frac{C\pi}{C}\}$. The contrastive loss now only needs one of its terms and can be written as

$$\mathcal{L}_D(x_1, x_2, y_p) = ||D(x_1, x_2) - y_p||_1^1. \tag{S4}$$

The other change that is needed is to apply the softmax at the output of the classifier, so that $y \in \Delta^C$.

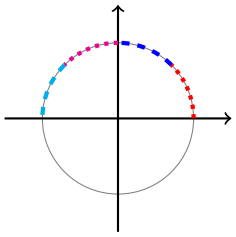


Figure S4: Hypothesis to separate multiple groups of data using the sinusoid projection. The example considers $C = 4$.

S6. Classification and statistics

The experiments compare, per cohort, the following views, each classified with an l_2-regularised logistic regression (Equation S1), a linear classifier that allows a direct assessment of representational discriminative power and easy interpretation [35]: (i) EEG-only views: the band-power view v_0 (relative log power); absolute log band power; and two views matched in information to the synthesized view, namely the temporal mean and standard deviation over pseudo-TRs of the synthesizers' own input spectra (57 channels $\times$ 97 bins, v_1^{ms}), and the same summary of a sparse random projection [20] to 8,192 dimensions of each pseudo-TR's complete $57 \times 97 \times 20$ input, an untrained linear map of exactly the information a synthesizer receives; (ii) the synthesized-fMRI view v_2 of each synthesizer, reduced to 20 principal components, and the same view from the untrained synthesizers; (iii) the combinations $v_0 \oplus v_2$ and $v_1^{\text{ms}} \oplus v_2$, each high-dimensional block reduced by its own 20 components; (iv) the contrastive sinusoid-separation classifier of Section S5 (two variants). Sensitivity analyses repeat the classification with 5, 10 and 40 components; with synthesized volumes average-pooled in blocks of $3 \times 4 \times 4$ voxels; with covariates regressed out of every feature within training folds (age and sex in every cohort, chlorpromazine-equivalent dose in the psychosis cohorts, recording hour in Semmelweis and Fribourg, recording year in Fribourg); and with the ICA-cleaned inputs and, in Parkinson's disease, the rest-only crop of Section S3. All transformations (standardisation, covariate regression, principal components) are fitted on training folds only; principal components of high-dimensional blocks are computed exactly from the subject Gram matrix. Synthesis ran on NVIDIA A100, H100 and L4 GPUs and the linear models on CPU nodes.

0.0.3 Validation

To validate each view v_i, we perform repeated stratified 10-fold cross-validation at the subject level (20 repeats; 3 for the contrastive classifier), with one recording per person, and select the regularisation strength $C \in \{10^{-3}, 10^{-2}, 10^{-1}, 1, 10\}$ by inner stratified 5-fold cross-validation within each training fold (Algorithm S1). We report the AUC averaged over repeats with a 95% bootstrap confidence interval, obtained by resampling subjects (2,000 stratified resamples) with their repeat-averaged out-of-fold probabilities, and a one-sided label-permutation p-value, $p = (1+\#\{\text{AUC}_{\text{perm}} \geq \text{AUC}\})/(1+N)$. Each permutation re-runs the complete nested procedure with two repeats; the null is therefore slightly wider than that of the 20-repeat estimate, which makes the test conservative. $N = 1{,}000$ for the EEG views and 500 for the synthesized views (minimum $p \approx 0.001$ and 0.002); for the contrastive classifier, whose every permutation retrains the network, $N = 20$ (minimum $p \approx 0.048$), and its p-values are treated as exploratory. The added value of synthesis is $\Delta\text{AUC} = \text{AUC}(v_0 \oplus v_2) - \text{AUC}(v_0)$, and likewise relative to v_1^{ms}. It is estimated with a paired stratified bootstrap over subjects (20,000 resamples), which gives a percentile 95% interval, percentile intervals at the Bonferroni levels and a two-sided bootstrap p-value; p-values are Holm-adjusted within the family of the 20 trained-synthesizer comparisons (four synthesizers, five cohorts) and, separately, within the 40 comparisons that include the untrained synthesizers. Leave-one-subject-out estimates are reported only as a sensitivity analysis: with weak features they are biased below chance, because the class balance of each training set mirrors the held-out label [36].

Below-chance estimate of CNN-TAG in FEP2

The below-chance AUC of CNN-TAG in FEP2 is not explained by a nuisance factor: none of its first 20 principal components was associated with sampling rate, recording duration, age, sex or medication dose ($|\rho| \leq 0.20$), and no subject carried more than 26% of any of the first five. In every outer fold the inner cross-validation AUC was below 0.5 (0.35–0.44) while the training AUC was 0.71–0.83, the signature of anti-learning in features without signal: when the full-sample difference between groups is close to zero,

Algorithm S1 Nested repeated cross-validation for view v_i

Require: individuals S with labels y; view v_i; hyperparameter grid $\mathcal{H}$; R repeats
Ensure: AUC, out-of-fold probabilities P

for $r = 1$ **to** R **do**
  for each stratified fold $(S_{\text{train}}, S_{\text{test}})$ of S **do** ▷ 10 folds, subject level
    fit standardisation (and PCA) on S_{train} only
    $h^\star \leftarrow \arg\max_{h \in \mathcal{H}} \mathrm{CV}_5(v_i, S_{\text{train}}, h)$ ▷ inner folds within S_{train}
    fit v_i with $h^\star$ on S_{train}; $P_r[S_{\text{test}}] \leftarrow v_i.\textsc{predict}(S_{\text{test}})$
  end for
end for
return $\frac{1}{R}\sum_r \mathrm{AUC}(P_r, y)$, $\frac{1}{R}\sum_r P_r$

the difference in a training fold is opposite to that in its held-out fold, so a fitted direction reverses out of sample [36].

S7. What the synthesizers learn

To interpret the classification results we characterise what the synthesizers learn on NODDI. (a) *Image protocol*: leave-one-subject-out over the 16 subjects (two validation subjects per fold for model selection), scored with SSIM, RMSE and voxelwise temporal correlation against an EEG-free population template (the mean training volume), with trigger-aligned EEG, EEG paired from the file start, and ECG only. (b) *Fair target*: BOLD motion-corrected (rigid registration, FSL), cleaned with the Friston-24 motion model and spike regressors for framewise displacement above 0.5 mm [12, 26], high-passed at 0.01 Hz, z-scored, smoothed (6 mm) and averaged into the 48 Harvard-Oxford grey-matter parcels present in every subject plus the global signal; ridge regression from lagged band power or full spectra, and UnEBOLT [22] with its pretrained EEG encoder [18], predicted the parcel time series of held-out subjects. Significance used circular-shift and cross-subject swap nulls on detrended predictions and targets, and IAAFT surrogates of the BOLD with model refitting [24, 33].

References

[1] Matthew A Albrecht et al. Time and frequency dependent changes in resting state EEG functional connectivity following lipopolysaccharide challenge in rats. *PLoS One*, 2018.

[2] Jimmy Lei Ba et al. Layer normalization. *arXiv*, 2016.

[3] Carla Barros et al. Advanced eeg-based learning approaches to predict schizophrenia: Promises and pitfalls. *Artificial intelligence in medicine*, 114:102039, 2021.

[4] Alan T Bates et al. Low-frequency EEG oscillations associated with information processing in schizophrenia. *Schizophrenia research*, 2009.

[5] Adrian P Burgess et al. Functional connectivity of gamma EEG activity is modulated at low frequency during conscious recollection. *International Journal of Psychophysiology*, 2002.

[6] David Calhas and Rui Henriques. Eeg to fmri synthesis for medical decision support: A case study on schizophrenia diagnosis. *medRxiv*, 2023. doi: 10.1101/2023.08.07.23293748.

[7] David Calhas et al. Eeg to fmri synthesis benefits from attentional graphs of electrode relationships. *MLHC*, 2023.

[8] James F. Cavanagh. EEG: 3-stim auditory oddball and rest in Parkinson's. OpenNeuro ds003490, 2021.

[9] Fani Deligianni, David W. Carmichael, Gary H. Zhang, Chris A. Clark, and Jonathan D. Clayden. NODDI and tensor-based microstructural indices as predictors of functional connectivity. *PLoS ONE*, 11(4):e0153404, 2016.

[10] Fani Deligianni et al. Relating resting-state fMRI and EEG whole-brain connectomes across frequency bands. *Frontiers in Neuroscience*, 2014.

[11] Andrew A Fingelkurts et al. Impaired functional connectivity at EEG alpha and theta frequency bands in major depression. *Human brain mapping*, 2007.

[12] Karl J. Friston, Steven Williams, Robert Howard, Richard S. J. Frackowiak, and Robert Turner. Movement-related effects in fMRI time-series. *Magnetic Resonance in Medicine*, 35(3):346–355, 1996.

[13] Dongyi He et al. Spec2VolCAMU-Net: A spectrogram-to-volume model for EEG-to-fMRI reconstruction. *Journal of Neural Engineering*, 2025. doi: 10.1088/1741-2552/ae15bf.

[14] Kaiming He et al. Deep residual learning for image recognition. In *CVPR*, 2016.

[15] Sibylle K Horat et al. Impaired social cognition in schizophrenia during the ultimatum game: An eeg study. *Schizophrenia research*, 2018.

[16] Turan M Itil. Qualitative and quantitative eeg findings in schizophrenia. *Schizophrenia bulletin*, 1977.

[17] Eric Jacobsen et al. The sliding DFT. *IEEE Signal Processing Magazine*, 2003.

[18] Wei-Bang Jiang, Li-Ming Zhao, and Bao-Liang Lu. Large brain model for learning generic representations with tremendous EEG data in BCI. *International Conference on Learning Representations*, 2024.

[19] Romeo Lanzino, Federico Fontana, Luigi Cinque, Francesco Scarcello, and Atsuto Maki. NT-ViT: Neural transcoding vision transformers for EEG-to-fMRI synthesis. In *ECCV Workshops*, 2024.

[20] Ping Li, Trevor J Hastie, and Kenneth W Church. Very sparse random projections. In *Proceedings of the 12th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 287–296, 2006. doi: 10.1145/1150402.1150436.

[21] Yamin Li et al. Neurobolt: Resting-state EEG-to-fMRI synthesis with multi-dimensional feature mapping. *Advances in Neural Information Processing Systems*, 2024.

[22] Yamin Li et al. UnEBOLT: A unified model for EEG-to-BOLD translation and functional connectivity reconstruction. In *Medical Imaging with Deep Learning (MIDL)*, 2026.

[23] Giulia Lioi, Claire Cury, Lorraine Perronnet, Marsel Mano, Elise Bannier, Anatole Lécuyer, and Christian Barillot. Simultaneous EEG-fMRI during a neurofeedback task, a brain imaging dataset for multimodal data integration. *Scientific Data*, 7:173, 2020. doi: 10.1038/s41597-020-0498-3.

[24] Neil Mehta et al. Time-varying EEG spectral power predicts evoked and spontaneous fMRI motor brain activity. *arXiv*, 2025.

[25] Anna Padée et al. "fribourg Ultimatum Game in Schizophrenia Study", 2022.

[26] Jonathan D. Power, Anish Mitra, Timothy O. Laumann, Abraham Z. Snyder, Bradley L. Schlaggar, and Steven E. Petersen. Methods to detect, characterize, and remove motion artifact in resting state fMRI. *NeuroImage*, 84:320–341, 2014.

[27] Frigyes Samuel Racz, Kinga Farkas, Melinda Becske, et al. Reduced temporal variability of cortical excitation/inhibition ratio in schizophrenia. *Schizophrenia*, 11:20, 2025. doi: 10.1038/s41537-025-00568-3.

[28] Ali Rahimi et al. Random Features for Large-Scale Kernel Machines. In *NIPS*, 2007.

[29] Brian J Roach et al. Event-related EEG time-frequency analysis: An overview of measures and an analysis of early gamma band phase locking in schizophrenia. *Schizophrenia bulletin*, 2008.

[30] Kristofer Grover Roos et al. From Brainwaves to Brain Scans: A Robust Neural Network for EEG-to-fMRI Synthesis. *arXiv*, 2025.

[31] Dean F. Salisbury et al. EEG: First episode psychosis vs. control resting task 1. OpenNeuro ds003944, 2021.

[32] Dean F. Salisbury et al. EEG: First episode psychosis vs. control resting task 2. OpenNeuro ds003947, 2021.

[33] Thomas Schreiber and Andreas Schmitz. Improved surrogate data for nonlinearity tests. *Physical Review Letters*, 77(4):635, 1996.

[34] Matthew Tancik et al. Fourier features let networks learn high frequency functions in low dimensional domains. *arXiv*, 2020.

[35] Marcos V Treviso et al. The explanation game: Towards prediction explainability through sparse communication. *arXiv*, 2020.

[36] Gaël Varoquaux. Cross-validation failure: small sample sizes lead to large error bars. *NeuroImage*, 180:68–77, 2018.

[37] Weiheng Yao, Zhihan Lyu, Mufti Mahmud, Ning Zhong, Baiying Lei, and Shuqiang Wang. CATD: Unified representation learning for EEG-to-fMRI cross-modal generation. *IEEE Transactions on Medical Imaging*, 2025. doi: 10.1109/TMI.2025.3550206.